What Changes a Mind? Length, Timing and Evidence on r/ChangeMyView

Group name	DataMindz
Dataset	r/ChangeMyView archive (raw coverage: 2013–2025; analysis window: Apr 2017–Nov 2022)
Code	`https://github.com/pranaybansal19/Data-Science-Project` (notebook `CMV_Analysis.ipynb` and README)

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1. Dataset

Source and coverage: the raw r/ChangeMyView (CMV) archive was obtained with the Arctic Shift Reddit Download Tool. It covers January 2013 through December 2025; this is the archive coverage, not the analysis period. After quality and consistency filters, all descriptive results and hypothesis tests use posts from 1 April 2017 through 30 November 2022. In CMV, an original poster (OP) may award a delta (Δ) to a reply that changes their view, and DeltaBot records the award in a confirmation comment.

Raw-data challenges: the archive has no ready-made outcome column: delta awards appear only inside DeltaBot comments. Many post and reply bodies are deleted or removed, schemas vary across years, 2,508 comment IDs are duplicated, and IDs change representation in 2023. Across the complete files we observed 132 post fields and 81 comment fields.

The notebook streams both compressed archives once and retains only the fields required for filtering, reconstruction, exploratory analysis and hypothesis tests. This avoids loading 18.6 million complete comment objects into memory.

	Raw files	After encoding
Instances (rows)	345,098 posts, 18,644,355 comments	479,276 replies
Attributes (columns)	132 / 81	11

Size check: $479{,}276 \times 11 = 5{,}272{,}036 \geq 3{,}000{,}000$. Each row is one eligible top-level reply. The 11 encoded attributes contain reply length and its log transform, response delay, five delay indicators, a link indicator, post length and the binary outcome.

Selected analytical features

Feature	Type	Real-world meaning
`won`	binary, reply	direct OP award, or a later OP award to the opening author's comment in the same branch
`words`	discrete, reply	whitespace-separated token count in the reply
`log_words`	continuous, reply	$\log(1 + \texttt{words})$, used for length inference
`delay_h`	continuous, reply	hours from the post to the reply
delay indicators	five binary columns	<1, 1–3, 3–6, 6–24 and >24 hours
`has_link`	binary, reply	text contains `http://`, `https://` or `www.`
`post_words`	discrete, reply	whitespace-separated token count in the original post
`op_awarded`	binary, thread	OP gave at least one delta; used for Test 4, not part of the 11-column reply table

2. Data preparation

Issue found	What we did	Why, and not the alternative
no outcome column	rebuilt it: DeltaBot "Confirmed" → the message it confirms must be by the OP and reply to a comment → that comment won	a Δ character can occur in quoted or instructional text; requiring an OP-authored award and bot confirmation reduces false labels
deltas often land on a later comment in a branch	the top-level reply wins if the OP awarded it, or a later comment by the same author in its branch	26,573 awards followed a back-and-forth; direct-award matching alone would mislabel those opening replies
inconsistent award capture outside the analysis period	kept posts from 1 Apr 2017 to 30 Nov 2022	this provides a stable observation window and avoids the sharp archive discontinuity visible after 2022 in Figure 1
mod posts, meta threads	kept titles starting with "CMV"	CMV's rules require it, so anything else is not a view
NSFW posts	dropped (1,326 posts)	their restricted visibility creates a different exposure process
post removed, or OP account deleted	thread dropped, not imputed	the text or OP identity required for measurement is unavailable; imputing argument text or authorship would be invalid
OP never awarded a delta	dropped for Tests 1–3, kept for Test 4	Tests 1–3 require at least one observed OP award; Test 4 explicitly compares award and non-award threads
deleted replies, OP's own, bots and mod notices	dropped	no text to measure, or not an argument
replies under 10 words	dropped (20,451)	an eligibility rule for substantive top-level arguments; it removes 4.1% of otherwise eligible replies, so results remain conditional on this threshold
2,508 duplicate comment IDs	kept the first copy	the same comment counted twice
skewed lengths, 5–6% beyond the IQR fence	kept, analysed on $\log(1+x)$	long replies are valid observations. The transform reduces skewness from 3.3 to 0.02 and lets differences be interpreted as ratios
delay is very skewed	binned (<1 h, 1–3, 3–6, 6–24, >24 h), one-hot	we compare rates by delay, never average it

Why this window: we retain posts from 1 April 2017 through 30 November 2022. Earlier records have less reliable bot-based award capture, while the series changes sharply after 2022. Within the selected period the quarterly share of intact threads with an OP award is comparatively stable (Figure 1), giving the four tests a more consistent measurement regime.

3. What changed after preparation

Measure	Before	After
Posts	345,098	44,160 intact CMV threads; 25,447 where the OP gave a delta
Comments → replies	18,644,355	479,276 top-level replies
Time span	raw archive: Jan 2013–Dec 2025	analyzed posts: 1 Apr 2017–30 Nov 2022
Outcome	none	`won`: 42,197 winners (8.8%)
Missing text	66% of posts removed	0
Duplicate IDs	2,508	0

The largest reduction occurs at thread level because valid post text and OP identity are required. Within retained threads, the pipeline removes 83,114 deleted replies, 29,204 bot or moderator notices, 166 OP-authored top-level comments and 20,451 replies below 10 words. Eight qualifying award threads have no reply remaining after these filters, so 479,276 analyzed replies represent 25,439 of the 25,447 qualifying threads. The final `won` field contains 42,197 positive outcomes.

4. Basic questions and problem statement

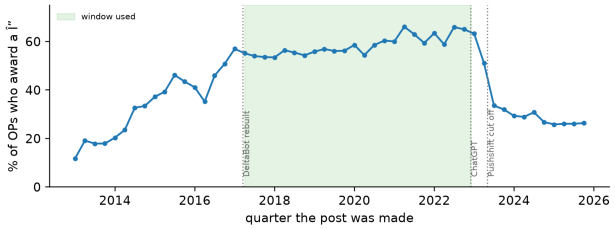


Figure 1: Quarterly share of intact CMV threads in which the OP awards a delta. The shaded region is the analysis window; dotted lines mark its boundaries and the later archive discontinuity.

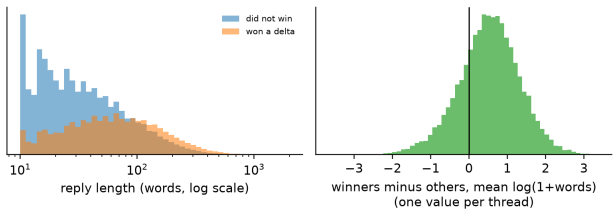


Figure 2: Reply length and outcome. Left: distributions for winning and non-winning replies on a log scale. Right: within-thread difference between winners and other replies; positive values mean winners are longer.

Basic questions

1. How long are replies, and how soon do they arrive? Median 84 words, 3.8 h after the post; 6.2% contain a link.
2. How many OPs award a delta? 57.6% of the 44,160 intact threads.
3. How many retained replies win? 42,197 of 479,276, or 8.8%. The observed rate declines as response delay increases.

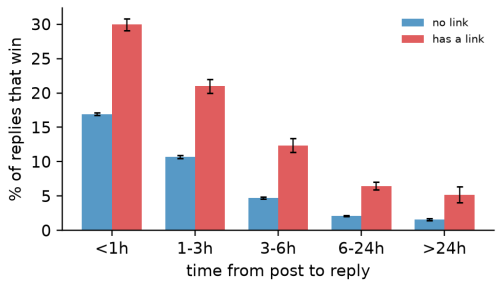


Figure 3: Win rate by response-delay bin and link status. Error bars show 95% confidence intervals. Earlier replies and replies containing a link have higher observed win rates.

Our problem statements

P1 (primary). Within the same thread, are the replies that win a delta longer than the ones that don't?

P2. Do replies posted within 3 hours of the post win more often than later ones?

P3. Do replies that link to a source win more often?

P4. Do OPs who explain their view at greater length award a delta more often, or less?

Analytical contribution: the pipeline reconstructs an explicit persuasion outcome from bot confirmations and reply trees. P1 controls for thread context through paired comparisons, P2 and P3 test timing and linked evidence, and P4 shifts the unit of analysis from replies to OP-authored threads.

5. Statistical inferences

Rows tested: Tests 1–3 use 479,276 retained replies from 25,439 represented threads, drawn from the 25,447 qualifying threads where the OP gave a delta. Test 4 uses all 44,160 intact threads. Every statistic is computed from the lecture formula and checked against `scipy`.

Practical-effect thresholds: for length comparisons we use a ratio of 1.20; for win-rate comparisons we use 2 percentage points, approximately 20% of the 8.8% base rate. Statistical significance is reported separately from whether an estimate crosses these practical thresholds.

Test 1 (P1): within a thread, are winning replies longer?

a) Hypotheses and test design

H_0	$\mu_d \le 0$
H_1	$\mu_d > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	paired t-test
Smallest difference that matters	winners 20% longer: $\mu_d \ge \ln 1.20 = 0.182$

For every eligible thread, d is the mean $\log(1 + \text{words})$ among winners minus the corresponding mean among other replies. The paired design compares replies judged in the same discussion and removes between-thread differences in topic and OP.

b) Other tests we considered

The paired t-test directly matches P1 and the dependence structure. The 23,055 thread-level differences are nearly symmetric (skewness -0.20), and the large sample makes inference on their mean stable.

Also considered	Why we did not use it
Welch t-test (unpaired)	ignores the thread, so busy threads full of short replies dominate. Run anyway: ratio 1.77, $t = 126.8$, same decision
Pooled t-test	also unpaired, and an F-test rejects equal variances ($F = 0.83$, $p < 10^{-100}$)
Two-sample z-test	needs the population σ, which we do not know
χ^2 on length bins	throws away the size of the gap and depends on where we cut

c) Result

Estimate	$\bar{d} = 0.519$: winners are 1.68× as long
95% confidence interval	1.66× to 1.70×
Test statistic (d.f.)	$t = 98.4$ (23,054)
p-value	$< 10^{-300}$
Decision	reject H_0

Within the same thread, winning replies are estimated to be 1.68 times as long as other replies. The entire interval exceeds the 1.20 threshold. This is a strong association, but it does not show that adding words alone causes persuasion.

Test 2 (P2): do early replies win more often?

a) Hypotheses and test design

H_0	$p_{\text{early}} - p_{\text{late}} \le 0$
H_1	$p_{\text{early}} - p_{\text{late}} > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	two-proportion z-test
Smallest difference that matters	2 percentage points

p_{early} and p_{late} are the winning proportions for replies posted within 3 hours and after 3 hours. The cutoff is close to the median delay and produces two large groups: 219,836 early and 259,440 late replies.

b) Other tests we considered

Both groups contain far more than 15 successes and failures, satisfying the large-sample condition for a two-proportion z-test.

Also considered	Why we did not use it
χ^2 independence, 5 delay bins	tells us the rate depends on delay, but not in which direction or by how much. Run anyway: $\chi^2 = 26{,}641$ (4), and the rate falls in every bin (Figure 3)
2×2 χ^2	the same test: $\chi^2 = z^2$
Welch t on delay	asks the reverse question (does winning predict delay?) and averages a very skewed variable
Paired t across threads	also valid, and it handles replies sharing a thread. Run as a check: +13.3 points, $t = 80.1$ (22,781), same decision

c) Result

Estimate	$+12.6$ points (15.6% early, 3.0% late)
95% confidence interval	$+12.4$ to $+12.8$ points
Test statistic	$z = 153.5$
p-value	$< 10^{-300}$
Decision	reject H_0

Early replies have a 12.6-point higher observed win rate and win about five times as often as later replies. The confidence interval lies well above the 2-point threshold. Timing remains observational and may also reflect visibility or reply quality.

Test 3 (P3): do replies with a link win more often?

a) Hypotheses and test design

$\boldsymbol{H_0}$	$p_{\text{link}} - p_{\text{none}} \leq 0$
$\boldsymbol{H_1}$	$p_{\text{link}} - p_{\text{none}} > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	two-proportion z-test
Smallest difference that matters	2 percentage points

The alternative tests whether replies containing at least one URL have a higher win rate. The link indicator measures the presence of a link, not the credibility or relevance of its destination.

b) Other tests we considered

The linked and unlinked groups contain 29,653 and 449,623 replies respectively, with enough successes and failures for the normal approximation.

Also considered	Why we did not use it
$2 \times 2\ \chi^2$	equivalent ($\chi^2 = z^2 = 3{,}645$), but two-sided only
Same z-test within length quartiles	not a replacement but a check, since linked replies are longer: the gap is 5.4 to 8.6 points in every quartile
Paired t-test	most threads have no linked reply to pair

c) Result

Estimate	$+10.3$ points (18.4% with a link, 8.2% without)
95% confidence interval	$+9.8$ to $+10.7$ points
Test statistic	$z = 60.4$
p-value	$< 10^{-300}$
Decision	reject H_0

Replies with a link have a 10.3-point higher observed win rate. The gap remains positive within every reply-length quartile and exceeds the practical threshold, although residual differences in topic, author effort and evidence quality may remain.

Test 4 (P4): do OPs with longer posts award more?

a) Hypotheses and test design

$\boldsymbol{H_0}$	$\mu_{\text{award}} - \mu_{\text{none}} = 0$
$\boldsymbol{H_1}$	$\mu_{\text{award}} - \mu_{\text{none}} \neq 0$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	Welch t-test
Smallest difference that matters	posts 20% longer (or shorter)

μ_{award} and μ_{none} are mean $\log(1 + \text{post words})$ for threads in which the OP did and did not award a delta. A two-sided alternative allows either group to write longer posts.

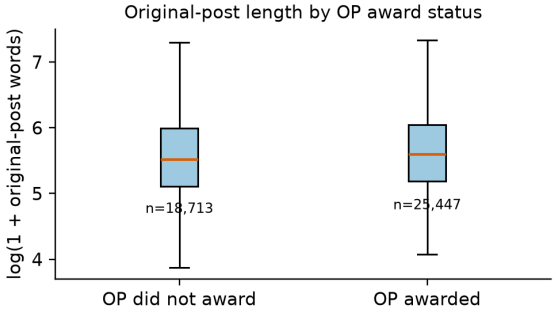


Figure 4: Test 4 comparison of original-post length by OP award status. Length is shown on the $\log(1 + x)$ scale used by Welch's t-test; outlier markers are hidden for readability, while all observations remain in the test.

b) Other tests we considered

The two groups are independent and their population variances are unknown. Welch's t-test estimates the mean difference without requiring equal variances.

Also considered	Why we did not use it
Pooled t-test	assumes equal variances, which the F-test rejects ($F = 0.955$, $p = 0.0007$). Run anyway: $t = 11.03$, same decision
Two-sample z-test	needs the population σ
Two-proportion z on award rate, long vs short posts	cuts length at an arbitrary point. Run anyway: 60.1% vs 55.2%
Paired t-test	each thread has one OP; nothing to pair

c) Result

Estimate	posts 1.07× as long (medians 270 vs 248 words)
95% confidence interval	1.06× to 1.08×
Test statistic (d.f.)	$t = 10.99$ (39,789)
p-value	4.5×10^{-28}
Decision	reject H_0

Awarding OPs write slightly longer posts on average, but the estimated ratio of 1.07 and its confidence interval remain well below the 1.20 practical threshold. The large sample detects a statistically clear difference that is small in magnitude.

6. Limitations

- **Association, not causation.** Reply length, timing and links may reflect effort, topic, visibility or argument quality; the tests do not isolate causal effects.
- **Dependence.** Replies share an OP and discussion context, so reply-level z-tests may understate uncertainty. Thread-level checks support the same directions.
- **Measurement and selection.** Deltas are public acknowledgements reconstructed from bot comments. Unreported changes of mind are invisible, and Tests 1–3 condition on intact threads with at least one OP award.
- **Short-reply threshold.** The 10-word rule removes non-substantive messages but is still a modeling choice. A full sensitivity analysis at 0, 5 and 10 words would show whether conclusions depend on it.
- **Time period.** Results describe April 2017 through November 2022 and may not generalize to later CMV activity.